

CRAIG LIONEL ROBERTS

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Sep 2024 – Dec 2026

MS in Artificial Intelligence; GPA: 3.81/4.0

Boston, MA

- **Relevant Coursework:** Machine Learning, Natural Language Processing, Algorithms, Linear Algebra and Probability for Data Science, Foundations of AI

VIT Bhopal University

Jul 2020 – May 2024

B.Tech in Computer Science and Engineering with AI and ML; GPA: 3.52/4.0

Bhopal, India

- **Relevant Coursework:** Data Structures, Database Management Systems, Discrete Mathematics, Foundations of Machine Learning

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++, Java, Git

LLM and Agent Systems: Multi-Agent Orchestration, LLM Supervisor and Router Architectures, Tool Use and Function Calling, OpenAI Agents SDK, Structured Output Extraction, Prompt Engineering, Guardrails

RAG and Retrieval: Retrieval-Augmented Generation, Hybrid Lexical and Vector Search, Azure AI Search, Azure OpenAI Embeddings, Vector Databases (Pinecone), Reranking, Sentence-BERT

Evaluation and Validation: LLM Evaluation, LLM-as-Judge, Pydantic Validation, MLflow Tracing and Evaluation

ML and DL: PyTorch, TensorFlow, scikit-learn, HuggingFace (Transformers, PEFT/LoRA), LLM Fine-tuning, BERT

Systems and Infrastructure: Databricks (Model Serving, Apps, Unity Catalog), FastAPI, asyncio, Azure

EXPERIENCE

Staples

Jan 2026 – Aug 2026

AI Engineer Co-op

Framingham, MA

- Architected a **5-agent document-extraction system** on **Databricks Model Serving** where an LLM supervisor classifies intent and routes cases to specialized agents run concurrently with **asyncio**, supporting 14 business action types across five domains.
- Engineered deterministic **Pydantic** validation over LLM output with bulk-request detection, enforcing mandatory fields and controlled values to return a stable, schema-validated JSON contract; shipped through **Azure DevOps CI/CD** across isolated dev, QA, and prod **Unity Catalog** environments.
- Built a constrained **RAG assistant** for customer-service reps on **Databricks Apps**, grounding answers in a corpus of **8,000+ approved knowledge articles** across 30 article groups through hybrid lexical and vector retrieval over **Azure AI Search** with Azure OpenAI embeddings and a GPT-4o-mini endpoint.
- Implemented a deterministic guardrail and **answer-normalization layer** with server-side entitlement filtering across 30 article groups, plus a **9-scorer offline evaluation framework** measuring citation accuracy, refusal, and latency over a labeled dataset.
- Designed a **multi-engine document-extraction service** normalizing Docling, Kreuzberg, and a Gemma 3 12B vision model into one validated schema, achieving **90% extraction accuracy** at **0.05s average latency** across 15+ file formats via a quality-gated fallback chain.

DeepRoot Minds Technologies

Jun 2023 – Dec 2023

Machine Learning Intern

Remote

- Fine-tuned pretrained **CNN** vision models via transfer learning on 3,000+ dermatological images, improving skin cancer classification performance by **12%** through targeted layer unfreezing and optimized preprocessing.
- Constructed a **GAN-driven** augmentation pipeline enhancing dataset diversity, increasing generalization capability by **10%** across evaluation sets, with **MLflow** experiment tracking.

PROJECTS

AdaptAd – Two-Agent Adversarial Ad Decision System (GitHub) (Live)

Jan 2026 – Apr 2026

- Developed a **two-agent adversarial system** (User Advocate vs. Advertiser Advocate) with opposing objectives, served through a **FastAPI** backend and React frontend with multi-provider LLM fallback (Groq, Gemini, template).
- Formulated ad delivery as an **optimization problem** solved with an **8-gene genetic algorithm** evolving decision thresholds, evaluated over engagement and CTR distributions from **MovieLens 25M** and **Avazu (40M rows)**.

Candidate Recommender System – Semantic Matching and Predictive Ranking (GitHub)

Aug 2025 – Sep 2025

- Assembled a **Sentence-BERT** embedding ranker fine-tuned on candidate-job pairs with contrastive learning, achieving **89% precision@5** across 100+ benchmark queries with sub-3-second scoring.
- Analysed failure modes across query distributions and iterated on training-data curation and loss formulation to improve ranking generalization.

NomadAI – RAG-Powered Travel Recommendation Engine (GitHub)

Jul 2025 – Aug 2025

- Assembled an end-to-end **RAG pipeline** indexing 2,400+ Reddit discussions into **Pinecone** via dense embedding retrieval, tuning chunk size, overlap, and top-k to balance recall against token budget before LLM synthesis.
- Integrated Reddit and Google Places APIs into a scalable ingestion pipeline and launched a **Streamlit** application serving context-driven, location-aware recommendations.